

Discriminatory Experiences, Critical Consciousness Development, and Well-Being Among Emerging Adults in and Beyond the First Year of the COVID-19 Pandemic

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As part of the developmental stage of emerging adulthood, youth may cultivate critical consciousness (CC) to transform oppressive systems. CC development may be influenced by discriminatory experiences and may affect well-being. To better understand longitudinal CC development and its relationship to discrimination and well-being (i.e., perceived stress, anxiety, hopefulness), we studied a U.S. national longitudinal cohort of emerging adult college students between the ages of 18 and 22 at baseline ($M_{\text{age}} = 20.0$, $SD = 1.3$) who completed four surveys between April 2020 and July 2021 ($N = 684$). The analytic sample was 63% women and 37% men (gender-diverse participants were removed due to small sample size) and 26% lesbian, gay, bisexual, queer/questioning, and other minoritized sexual identities. Self-identified race/ethnicity backgrounds were 54% white, 20% Asian/Pacific Islander, 9% Latinx, 5% Black, and 10% multiple races/ethnicities and/or as Middle Eastern/North African. We conducted latent profile transition analysis and identified five patterns of CC development, with a minority of participants in two “growth” transition patterns wherein they increased their CC over time. Maintaining higher levels of CC, and especially developing CC, was associated with more prior experiences with discrimination and with concurrent and subsequent higher levels of perceived stress and anxiety. We recommend institutions of higher education and college-based organizations to build well-being practices and structures into CC-raising spaces to support empowering CC development amid ongoing sociopolitical turmoil.

Public Significance Statement

As youth develop their critical consciousness (awareness of and actions to dismantle structural oppression) in response to sociopolitical events like mass protest movements and national elections, they may require support for their well-being. Among an emerging adult college student sample surveyed during the tumultuous sociopolitical events of 2020 and 2021, those who developed their critical consciousness reported greater prior experiences of discrimination and greater concurrent and subsequent experiences of stress and anxiety.

Keywords: critical consciousness, sociopolitical development, well-being, mental health, emerging adults

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continued

Emerging adults ages 18–29 (Arnett, 2023) experienced the first year of the COVID-19 pandemic during a critical developmental period. As they contended with magnified, intersecting injustices arising from the pandemic and other social forces, they may have developed critical consciousness (CC), defined as the capacities of reflection, motivation, and action against structural inequities (Diemer et al., 2022; Watts et al., 2003). Specifically, emerging adults may have developed CC in response to injustices including rapid increases in anti-Asian racism, a global surge in the Black Lives Matter (BLM) movement; increases in antilebian, gay, bisexual, transgender, queer/questioning, and other minoritized sexual and gender identities legislation; and a polarized U.S. national election (Home et al., 2022; Karras et al., 2022). For emerging adults facing societal marginalization, experiences of discrimination by race/ethnicity, gender, and/or sexuality in this sociopolitical context may have catalyzed CC development (Karras et al., 2022; Moffitt et al., 2024; Wilf et al., 2023). For emerging adults experiencing societal privilege based on their racial, gender, and/or sexual identities, observing discrimination and the resultant social movements (e.g., BLM, Stop Asian Hate) may have catalyzed new thinking and action against inequity (Karras et al., 2022; Moffitt et al., 2024; Quiles et al., 2023). Many emerging adults experienced increased threats to their well-being during the pandemic, including increased rates of anxiety and stress (Hoyt et al., 2021, 2023) and decreased levels of hopefulness (Genç & Arslan, 2021). Prior research suggests that emerging adults may experience increased well-being challenges in relation to their CC, especially their critical action (Maker Castro et al., 2022b). It remains unclear what CC development over time (and in the context of injustices characterizing the year 2020) looked like and how any such development may have related to emerging adults' well-being. This study uses four waves of longitudinal data collected from college students between April 2020 and July 2021 and is, to our knowledge, the first quantitative study to longitudinally assess CC development in relation to well-being and experiences of discrimination among emerging adults.

CC in a Sociopolitical Development Framework

CC development represents how the distinct dimensions of critical reflection (seeing and analyzing social inequities), critical motivation (feeling interested and able to act against inequities), and critical action (efforts to challenge inequities) evolve. CC dimensions may grow reciprocally (Watts et al., 2003), or youth may develop more along one dimension of CC than others (Godfrey et al., 2019). Sociopolitical development (SPD) theory conceptualizes five patterns or stages of CC development: acritical (no awareness of inequity), adaptive (growing awareness), precritical (becoming concerned about inequities), critical (motivated to take action), and liberated (acting to support social change; Watts et al., 2003). While SPD scholars originally moved away from stages to develop more process models (Watts & Guessous, 2006), the stage model

can highlight the potential nonlinear and cyclical SPD developmental processes and identify where youth are across the CC dimensions (Hope et al., 2023; Watts et al., 2003). By accounting for nonlinearity in the stage model, we can further consider how different constellations of CC dimensions may be differentially catalyzed by experiences of discrimination (e.g., Do discriminatory experiences prompt increased analysis of injustice without corresponding action?) and how the development of these CC constellations may affect well-being.

The Role of Marginalization and Privilege in CC Development

Pathways to CC (and the subsequent relationships to well-being) are likely shaped by youth's multiple social locations with systems of privilege and marginalization (e.g., by race/ethnicity, gender, sexuality; Hope et al., 2018; Leath et al., 2022). We conceptualize marginalization as a multidimensional and multi-contextual "web of processes" that generate systemic power imbalances between groups (Causadias & Umaña-Taylor, 2018, p. 709). Discrimination, or unequal treatment of a person or group at the individual and/or structural level due to their social location(s), is one way to operationalize experiences with marginalization and may inform CC development (Anyiwo et al., 2018; Fine et al., 2018). Experiencing discrimination may catalyze understanding and challenges to the systems that perpetuate these inequalities (Anyiwo et al., 2018). We conceive privilege as an experience wherein youth benefit from the web of systems and structures perpetuating systemic inequities (e.g., heteropatriarchy, systemic anti-Black racism). We conduct exploratory analyses of how youth may differ in CC based on social locations within systems of privilege and marginalization (i.e., by race/ethnicity, gender, and sexuality).

Marginalization and Privilege by Race

Youth of color and white¹ youth have distinct developmental pathways toward CC given their differing positions within society's racial hierarchy (Wray-Lake et al., 2023). Youth of color's experiences with systemic and interpersonal racism have been found to inform CC development (Anyiwo et al., 2018; Hope et al., 2023). Youth of color, including Black, Latinx, Asian, and Indigenous youth, endured disproportionate effects of the COVID-19 pandemic (Tai et al., 2021). Amid racialized health disparities, they may also have experienced collective forms of racial trauma stemming from the globalized killings of Black Americans, rising anti-Asian discrimination, and ongoing xenophobic rhetoric and policy targeting Latinx and Black immigrants

¹ We do not capitalize white or any of its derivatives as we recognize and challenge white people's racial identity as rooted in white supremacy (Sullivan, 2019).

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(Karras et al., 2022; Quiles et al., 2023). Youth of color may have responded with forms of critical action to advocate on behalf of inequities facing their racial groups and/or in solidarity with other racially marginalized groups (Karras et al., 2022; Wilf et al., 2023).

Meanwhile, CC development is vital for white youth to identify their racial privilege and encourage critical action (Hazelbaker et al., 2022). However, research suggests that white emerging adults experience difficulty deepening their critical reflection on racial injustice (Dull et al., 2022). In a (Hazelbaker et al., 2022) white supremacy society, white youth experience the pervasive belief that they are the cultural norm and they may fail to engage in antiracism due to the ways they benefit from power structures (Jardina, 2019). The racially motivated protests in 2020 may have challenged inclinations of color evasion (i.e., ignoring racialization) and power evasion (i.e., avoiding confrontation with institutional racism; Yi et al., 2023). Within the context of 2020, white youth engaged in critical action while hesitating to discuss their racial positions of privilege (Moffitt et al., 2024; Quiles et al., 2023). Thus, their CC development may emerge distinctly from their peers of color, who were more likely during this period to engage in sociopolitical events with a more developed critical reflection, including related to their racialized identities and experiences.

Marginalization and Privilege by Gender and Sexuality

Emerging adults who identify as women; lesbian, gay, bisexual, queer/questioning, and other minoritized identities (LGBQ+); and/or transgender and gender diverse may develop CC in response to their positions within marginalizing systems (Conner et al., 2023; Fine et al., 2018). LGBQ+ youth (Maker Castro et al., 2022a) and women (Ballard et al., 2020; Hope et al., 2022; Maker Castro et al., 2022a) report higher levels of CC than straight and male counterparts. Experiences with gender and sexuality within the sociopolitical context of 2020 may have informed CC. Emerging adult women experienced ongoing efforts to undermine reproductive rights (Brysk, 2023), inequities in income and governmental leadership (England et al., 2020; Misra, 2020), and a national reckoning against sexual violence (Burke, 2018). Women responded with mass mobilizations, including the March on Washington (Martin & Smith, 2020) and the #MeToo movement (Burke, 2018). Attacks on the lesbian, gay, bisexual, transgender, queer/questioning, and other minoritized sexual and gender identities community included institutional efforts to prevent same-sex adoptions and curb health care for sexual and gender minorities (Horne et al., 2022). Lesbian, gay, bisexual, transgender, queer/questioning, and other minoritized sexual and gender identities youth responded with ongoing resistance to homophobia in their schools and communities (Robinson & Schmitz, 2021). Men and straight youth may have developed CC in ways similar to the ways in which white youth may have developed their CC within the sociopolitical context of 2020: They may have experienced social movements for gender and sexuality equality that newly sparked their analysis and action against the heteropatriarchy. Alternatively, or additionally, those who are white and men/straight may have used their developing CC of systemic racism as an entry point into considering other marginalizing systems (Maker Castro et al., 2024).

The Relationship Between CC and Well-Being

Well-being is a multidimensional construct inclusive of mental, socioemotional, and physical health and both positive manifestations of each health dimension (e.g., hopefulness) and challenges across these dimensions (e.g., anxiety and stress; J. Johnson & Wood, 2017). Youth may encounter challenges to their well-being as they pursue CC (Hope et al., 2023; Maker Castro et al., 2022b). Prior researchers have operationalized anxiety and stress as well-being challenges related to CC development for emerging adults (e.g., Ballard et al., 2020; Hope et al., 2018; Maker Castro et al., 2022a). Hope et al. (2018) found that critical action exacerbated the negative effect of discrimination on mental health for Black youth but served as a protective mechanism for Latinx youth. Conversely, Ballard et al. (2020) found that critical action was negatively associated with well-being for Latinx youth, as well as white, Asian, and Hawaiian/other race youth, but no significant effect for Black youth. Finally, Maker Castro et al. (2022a) found that each of the three CC dimensions was associated with higher anxiety during the onset of the COVID-19 pandemic for a national sample of U.S. emerging adult college students. Heightened levels of stress and anxiety related to CC may be explained by experiences such as burnout from critical action and contending with institutions that resist youth-led action (Fernández et al., 2018; Vaccaro & Mena, 2011).

Emerging adults may conversely experience a stronger sense of well-being as a result of their CC (French et al., 2020). For instance, CC may occur in tandem with hopefulness in one's ability to make change (Christens et al., 2018). CC may also be able to help counteract other forces that affect well-being, like experiences of discrimination (Anyiwo et al., 2018), which can cause increases in anxiety and stress among emerging adult college students marginalized based on race/ethnicity, sexuality, and/or gender (Barrita et al., 2023). Discrimination may also deter the development of hopefulness among marginalized groups (Mitchell et al., 2020). Thus, we examine CC development within the context of well-being challenges (i.e., anxiety, stress) and assets (i.e., hopefulness) and in relation to experiences of discrimination.

Person-Centered Approaches to CC Development

Person-centered approaches acknowledge the complexity of CC as a developmental phenomenon (Godfrey et al., 2019; Sameroff & Mackenzie, 2003) by identifying subgroups within a population whose CC dimensions operate similarly. Through longitudinal person-centered approaches like latent profile transition analysis (LPTA), we can examine CC development by observing how subgroups may move, or stay, within profiles over time. We can also examine how distinct CC transition patterns may be related to experiences with discrimination and well-being (Anyiwo et al., 2018; Schwarzenthal et al., 2024).

Three recent cross-sectional investigations leveraged person-centered approaches to identify CC profiles among adolescents (Briggs et al., 2023; Godfrey et al., 2019; Schwarzenthal et al., 2024) and found similar patterns across distinct ethnic/racial samples, including Black adolescents (Briggs et al., 2023), African American, Chinese, Dominican, Puerto Rican, and other racial/ethnic minoritized youth (Godfrey et al., 2019), and ethnically racially diverse German youth from European, African, Middle Eastern, Asian, Australian, and

American descent (Schwarzenthal et al., 2024). These studies all assessed critical reflection and critical action; Briggs et al. (2023) and Godfrey et al. (2019) also assessed critical motivation. All studies identified youth at the precursor stages of SPD (i.e., *acritical*, meaning youth saw the world as fair and did not engage in critical action, or *adaptive*, meaning they were beginning to understand oppression and demonstrated little to no critical action). Godfrey et al. (2019) found that youth in the acritical profile had better well-being than more critically conscious peers. All studies also identified a precritical phase, the phase considered the beginning of CC wherein youth question unjust systems but may not engage consistently in action, suggesting that reflection can emerge ahead of motivation and action. Youth in this precritical phase reported lower well-being (Godfrey et al., 2019; Schwarzenthal et al., 2024). Finally, all studies identified youth in a critical or liberated stage of SPD. Schwarzenthal et al. (2024) found that participants with the highest levels of CC reported high levels of socioemotional well-being.

Two studies also examined the relationship between CC profile membership and experiences with ethnic/racial discrimination, with mixed results. Briggs et al. (2023) found no relationship between experiences of racial discrimination and CC profile membership, whereas Schwarzenthal et al. (2024) found that members of the “armchair activist” and “actionist” profiles reported more discrimination.

The Present Study

Emerging adults become increasingly independent and agentic as they form their trajectories into adulthood (Arnett, 2023), making this an important time for potential CC development (Wray-Lake & Ballard, 2023), including the 62% of high school graduates who enroll in postsecondary education (Hanson, 2024). As college students, emerging adults may develop their CC through academic coursework and student organizations that offer emerging adults opportunities to analyze structural oppressions or inspire critical action (Maker Castro et al., 2024). Our study uses LPTA, a longitudinal person-centered approach that can highlight heterogeneity in CC development. First, we aimed to identify CC profiles and examine how emerging adults transitioned through profiles over the year 2020. In line with extant cross-sectional latent profile studies, we hypothesized that we would identify at least three different profiles (i.e., acritical, precritical, and critical; Briggs et al., 2023; Godfrey et al., 2019; Schwarzenthal et al., 2024). Given that research suggests emerging adults in college engaged in CC in new ways during the year 2020 (Moffitt et al., 2024; Quiles et al., 2023; Wilf et al., 2023), we hypothesized that some participants would develop their CC (e.g., move from precritical to liberated) over time. Second, we aimed to investigate the relationships between the transition patterns and experiences with discrimination in April 2020 and well-being (i.e., perceived stress, anxiety, and hopefulness) at the time of the transitions across CC profiles in July 2020 and 1 year later in July 2021. We hypothesized that experiences of discrimination in April 2020 would be associated with transitions into more developed CC profiles in July 2020 (Anyiwo et al., 2018; Schwarzenthal et al., 2024). We further hypothesized that those who transitioned into more developed CC profiles would experience heightened perceived stress and anxiety at the time of transition given that prior research identifies moments of CC engagement as

challenging for emerging adult college students’ well-being (Maker Castro et al., 2022b). For those who maintained low levels of CC, we hypothesized lower levels of stress and anxiety (Godfrey et al., 2019), while we conversely hypothesized that those who maintained high levels of CC would experience heightened hopefulness across time, as sustained CC may be related to youths’ hope that they can make change (Christens et al., 2018). Third, we also explored preliminary relationships between transitions and demographic variables (i.e., race/ethnicity, gender, sexuality). Beyond conceptualizing demographic variables as proxies for experiences of marginalization (e.g., racism, sexism, heterosexism), we did not hypothesize specific roles of demographic variables given the exploratory nature of the investigation.

Method

Procedure

Participants were recruited with Instagram ads that targeted diverse emerging adult college students at the onset of the COVID-19 pandemic (late April 2020; T1) to share about their pandemic-related experiences (see Cohen et al., 2020 for details). Participants completed the survey via Qualtrics until the budget capacity for incentives of \$10 gift cards was met; after validity checks of college email and age (Hoyt et al., 2021), the sample size was reduced to 707 students from 49 states and Washington DC who attended 370 universities in April 2020 (Hoyt et al., 2021). Because we planned to explore categorical gender variables, we dropped 23 gender-diverse participants from the sample due to the small sample size (Cameron & Stinson, 2019); we encourage future studies to sample for greater gender diversity (Restar et al., 2021). Our final analytic sample was $n = 684$. Participants were invited by email to complete follow-up surveys via Qualtrics in July (T2; $n = 543$), November 2020 (T3; $n = 461$), and July 2021 (T4; $n = 339$) and received gift cards of \$5 at T2–T4 as honoraria. Because of resource constraints, not all survey measures were included at each time point (e.g., we asked about experiences of discrimination at T1 and T4 only).

Participants

The sample of $n = 684$ includes participants from a national cohort of emerging adult college students (Hoyt et al., 2021) between the ages of 18 and 22 ($M = 20.0$, $SD = 1.3$). The sample at T1 included 54% white, 21.1% Asian, 10.1% multiple races/ethnicities, 8.8% Latinx, 5.3% Black, and 1.1% Middle Eastern/North African participants. The sample included 63.0% women and 37.0% men, including transgender men, who were included to be inclusive of all male identities (Rogers, 2019). There were no transgender women participants. At T1, 26.6% of participants identified as LGBTQ+. At T1, 41.7% of participants were of immigrant origin (i.e., born outside of the United States or having one or both parents born outside of the United States). There were also variations in year in college and household income. Table 1 reports the sample composition at each timepoint, and Appendix A provides further description of sociodemographic variables. Retention rates were 78.4% at T2 ($n = 543$), 67.4% at T3 ($n = 461$), and 49.6% at T4 ($n = 339$). See Appendix B for missing data procedures and results.

Table 1*Demographic Profile of the Analytic Sample for Each Wave*

Measure	Time 1		Time 2		Time 3		Time 4	
	<i>N</i>	%	<i>N</i>	%	<i>N</i>	%	<i>N</i>	%
Number of participants	684		543		461		339	
Gender								
Women	431	63.0	343	63.2	302	65.5	235	69.3
Men	253	37.0	178	32.8	139	46.0	104	44.3
Sexuality								
LGBQ+	182	26.6	137	25.3	116	25.2	101	29.8
Straight	502	73.4	384	70.7	325	70.5	238	70.2
Race/ethnicity								
White	369	54.0	274	50.5	237	51.4	182	53.7
Black	36	5.3	20	3.7	17	3.7	13	3.8
Latinx	60	8.8	53	9.8	41	8.9	34	10.0
Asian	144	21.1	119	21.9	96	20.8	74	21.8
Multiple racial groups and Middle Eastern/North African	75	11.0	55	10.1	50	10.9	36	10.6
Immigrant origin								
Immigrant origin	285	41.7	232	42.7	250	54.2	189	55.8
Nonimmigrant origin	399	58.3	289	53.2	191	41.4	150	44.3
Socioeconomic position								
<\$26,000	88	12.9	70	12.90	52	11.3	40	11.8
\$26,000–\$53,999	143	20.9	113	20.81	99	21.5	71	20.9
\$54,000–\$99,999	190	27.8	150	27.62	127	27.6	98	28.9
\$100,000–\$249,999	220	32.2	157	28.91	135	29.3	110	32.4
>\$250,000	41	6.0	31	8.1	28	6.1	20	5.9

Note. Race/ethnicity was operationalized by creating five mutually exclusive racial/ethnic groups (i.e., white, Black, Asian, Latinx, and multiple racial/ethnic groups). We included the small group ($n = 8$) of Middle Eastern/North African participants with the multiple racial/ethnic group participants due to small sample size. We collapsed gender responses into three categories: man, woman, and transgender and gender diverse. Due to insufficient statistical power, we excluded gender-diverse and nonbinary youth from analyses. Sexual orientation was operationalized as a binary variable: straight/heterosexual or lesbian, gay, bisexual, queer/questioning, and other minoritized identities. Socioeconomic position was operationalized by estimating the participant's family's typical annual household income pre-COVID-19 pandemic. LGBQ+ = lesbian, gay, bisexual, queer/questioning, and other minoritized identities

Measures

Critical Consciousness

We used the 13-item Short Critical Consciousness Scale (Diemer et al., 2022) representing three dimensions: critical reflection (four items, e.g., “certain racial or ethnic groups have fewer chances to get good jobs”), critical motivation (four items, e.g., “it is important to correct social and economic inequality”), and critical action (five items, e.g., “participated in a civil rights group or organization”). Items are scored using Likert scales ranging from 1 (*strongly disagree*) to 6 (*strongly agree*) for critical reflection and motivation and 1 (*never did this*) to 5 (*at least once a week*) for critical action. Each subscale was independently averaged and analyzed upon satisfactory confirmatory factor analysis results. The Short Critical Consciousness Scale was administered in T1–T3. Cronbach's α s were .91, .94, and .93 for reflection; .77, .84, and .85 for motivation; and .84, .78, and .87 for action at T1–T3.

Anxiety

We used the General Anxiety Disorder seven-item (Spitzer et al., 2006). Each item was rated on a 4-point scale (0 = *not at all*, 3 = *severely, it bothered me a lot*). A total score was created by summing across items. The scale has been validated with college-age populations (Byrd-Bredbenner et al., 2020). In this sample, Cronbach's α was .91 at T2 and T4.

Perceived Stress

We used the 10-item Perceived Stress Scale (Cohen & Williamson, 1988). Each item was rated on a 5-point scale (0 = *never* to 4 = *very often*). Four items were reverse coded, such that higher scores indicated higher perceived stress. A total score was created by summing across items. The scale has been validated with college-age populations (Roberti et al., 2006). Cronbach's α s were .88 and .84 at T2 and T4. The T4 survey contained an error wherein Option 2 (*sometimes*) was not included. To address this error, responses were reassigned: *never* = 0, *almost never* = 1, *fairly often* = 3, and *very often* = 4. The revised scaling still ranges from 0 to 40, thus aligning with the correct version of the measure used at T2. As a sensitivity check, we rescaled baseline answer choices as follows: 0 = 0; 1 = 1.333; 2 = 2.66; and 3 = 4s, to ensure that results were robust to this data collection error (Hoyt et al., 2021).

Hopefulness

The Individual-Differences Measure of Hope (Snyder et al., 1991) included two subscales (agency and pathways). The survey used eight items (four items per subscale); items were scored from 1 = *definitely false* to 8 = *definitely true*. Studies with emerging adults demonstrated acceptable internal consistency ($\alpha = .89$; Snyder et al., 1991). Cronbach's α s for hope agency were .83 and .88 at T2 and T4. Cronbach's α s for hope pathways were .85 and .83 at T2 and T4.

Discrimination

The six multiple choice items from the Everyday Discrimination Scale (Williams et al., 1997) were used at T1, and the scale was slightly modified to address the specific nature of the COVID-19 pandemic. The modified prompt stated: “In your day-to-day life since the COVID-19 pandemic began, how often have any of the following things happened to you?” Items 1–6 are scored on a 5-point Likert scale (0 = *never*, 4 = *almost every day*). Cronbach’s α for items 1–6 was .86 at T1.

Analytic Approach

Overall Approach

We conducted analysis in Mplus 8.4 (Muthén & Muthén, 1998–2017). First, we used participants’ scores on the three scale scores for each CC dimension to create a latent profile analysis (LPA) model that assigns each participant a probability of belonging to different subgroups—or profiles—of CC dimension scores (Masyn, 2013). Next, we used latent profile transition analysis (LPTA) to connect the cross-sectional LPA models in a longitudinal mixture model that incorporated transition probabilities to describe participants’ movement between profiles over time points. Then, we used a manual three-step process (Asparouhov & Muthén, 2014) to test the relationship between longitudinal patterns of CC and well-being, discrimination, and demographic variables by adding these variables each as auxiliary variables to the unconditional LPTA model previously constructed. De-identified data and Mplus code are available upon reasonable request. The study and analyses were not preregistered.

Conducting Latent Profile Analysis

We ran LPA at each time point to capture heterogeneity in participants’ CC scores with a mixture model. To identify the best fitting model at each time point, models with different numbers of profiles were compared within four variance–covariance matrix specifications: profile-invariant diagonal, profile-varying diagonal, profile-invariance unrestricted, and profile-varying unrestricted variance–covariance specifications (S. K. Johnson, 2021). Evaluation of model fit is described in Appendix C.

Conducting the Unconditional Latent Profile Transition Analysis

The best fitting LPA models from each time point were linked together with transition probabilities to construct the unconditional LPTA model. We used the Bayesian information criteria (BIC) and sample size adjusted BIC indices to examine improvement in model fit between a constrained model (i.e., the means for each CC dimension were constrained to be equal across time for analogous profiles) and a model with freely estimated means (Nylund, 2007), with lower values in the constrained model indicating better fit over the freely estimated model (S. K. Johnson, 2021).

Three-Step Approach to Add Auxiliary Variables

We used a three-step process (Asparouhov & Muthén, 2014) for adding auxiliary variables to an unconditional LPTA to assess relationships between participants’ transition patterns and experiences

with discrimination and well-being and to explore preliminary relationships between transition patterns and demographic variables. The three-step approach, which must be completed manually (i.e., the software does not automatically implement all steps), allows for investigating how auxiliary variables related to each transition profile while holding the measurement model constant (Nylund-Gibson et al., 2014). Step 1 of the three-step process was determining the unconditional LPTA. In Step 2, nominal variables were created that capture most likely transition patterns using posterior probabilities from the unconditional LPTA, and the classification uncertainty is separately quantified. In Step 3, participants’ most likely transition pattern was regressed on auxiliary variables while fixing the classification uncertainty quantified in Step 2 (Asparouhov & Muthén, 2014). This produced a mean score for each auxiliary variable across transition patterns in the case of continuous auxiliary variables. In the case of categorical auxiliary variables (i.e., our demographic variables), the final step was a binary or multinomial logistic regression. Wald χ^2 tests were used to compare mean scores of the auxiliary variables between transition patterns within each time point and to compare mean scores over time. Due to multiple comparisons, we adjusted the family-wise error rate using the Holm–Bonferroni method (Holm, 1979).

Handling of Missing Data

Data were missing because of attrition and partial survey completion. Based on available missing data techniques for LPTA in Mplus, we used a combination of full information maximum likelihood (Enders & Bandalos, 2001) for indicators (CC variables) and multiple imputation for auxiliary variables (i.e., anxiety, perceived stress, hopefulness at T2 and T4; Little, 2013). As there is not yet clarification on how to conduct multiple imputation at the profile enumeration phase (S. K. Johnson, 2021), we used full information maximum likelihood when constructing the LPA models and the unconditional LPTA model to account for missing data due to attrition. We used multiple imputation, an effective way to reduce bias in estimates due to attrition (Little, 2013), for the LPTA with auxiliary variables because the three-step process defaults to listwise deletion on outcome variables. We imputed 50 data sets for each wave for the well-being variables using information from all other variables in the model. These 50 data sets were combined during the third step of the three-step process.

Measure of Model Precision

There are currently no specific procedures or guidelines for determining whether one has adequate sample size in LPTA (S. K. Johnson, 2021; Nylund-Gibson et al., 2023). Therefore, we examined whether our model had sufficient precision using an absolute (instead of relative) measure of model precision called the odds of correct classification (Masyn, 2013; Nylund-Gibson et al., 2023). Odds of correct classification estimate the ratio of the odds of correctly classifying participants compared with random assignment; the ratio should be >5 (Nagin & Odgers, 2010).

Transparency and Openness

This study was not preregistered. The data and analytic code necessary to reproduce the analyses presented here are available for

appropriate use upon emailed request to the corresponding author. Study procedures were approved by the Fordham University's Institutional Review Board, and all participants consented to their participation.

Results

Preliminary Analyses

Preliminary analyses included item-level descriptive statistics (Appendix D), longitudinal measurement invariance testing (Appendix E), descriptive statistics of study variables (Appendix F), and correlations between variables at each time point (Appendix G). The nonnormality of variables was addressed in Mplus using the robust maximum likelihood estimator (Kline, 2011).

Latent Profile Analysis

Model fit indices for the LPA at each time point are presented in Appendix H. At each time point, up to seven models per variance-covariance structure were run, and converged models are reported. Models with 7+ profiles contained profiles that were both very similar substantively and very small in sample size, suggesting overextraction of profiles. The candidates for final models within each time point were chosen based on entropy and the model fit indices of Akaike information criteria, BIC, and sample size adjusted BIC. When there were more than one candidate models with similar model fit indices and entropy values, we looked to theoretical alignment and classification precision (such as odds of correct classification; see Appendix I). The final models chosen (presented in bold in Appendix H) were the four-profile class-invariant diagonal model at T1 and the four-profile profile-invariant unrestricted model at T2 and T3.

Table 2 reports the number of participants classified into the profile they are most likely to be in. Figure 1 presents profile z scores. The profiles were named in alignment with previously identified CC latent profiles (Briggs et al., 2023) and SPD theory (Watts et al., 2003). *Acritical bystanders* were characterized by low levels of endorsements across the three CC domains when compared with the mean levels of CC across participants. *Precritical bystanders* were characterized by CC scores very similar to the overall average for CC dimensions for the full sample, with reflection and motivation scores that were higher than action scores. *Liberated actors* were characterized by high scores across the three CC dimensions when compared with the sample average score. *Acritical actors* were characterized by high levels of action and lower reflection and motivation compared with the sample average. This profile reflected a constellation of CC dimensions not previously theorized in SPD theory (Briggs et al., 2023; Watts et al., 2003). We conducted exploratory analyses (Appendix J and K) because we did not identify any longitudinal transition patterns that included acritical actors due to the small sample size.

Latent Profile Transition Analysis

We found the constrained three-time, four-profile LPTA model to have lower BIC and sample size adjusted BIC over the freely estimated model (Appendix L; S. K. Johnson, 2021). The transition probability matrix is presented in Table 2 and depicted using a Sankey diagram in Figure 2. With four profiles over three time points, there were 64 potential transition patterns (e.g., Profile 1 → Profile 1 → Profile 1, Profile 1 → Profile 1 → Profile 2, and so on). We explored our second and third aims with the five most prevalent transition patterns (i.e., the five transition patterns with the highest likelihood of membership).

Table 2

Latent Profile and Transition Probability Matrix of Four-Profile Models

Latent profile prevalence	Acritical bystander	Acritical actor	Precritical bystander	Liberated actor
Time 1 (April 2020)	179	27	356	122
Time 2 (July 2020)	113	44	383	144
Time 3 (November 2020)	108	59	377	139
Transition probability (T1 → T2)				
Time 1 latent profile (April 2020)	Time 2 latent profile (July 2020)			
	Acritical bystander	Acritical actor	Precritical bystander	Liberated actor
Acritical bystander ($n = 179$)	61%	12%	28%	0.0%
Acritical actor ($n = 27$)	18%	44%	27%	11%
Precritical bystander ($n = 356$)	0.0%	2%	89%	9%
Liberated actor ($n = 122$)	0.0%	3%	9%	88%
Transition probability (T2 → T3)				
Time 2 latent profile (July 2020)	Time 3 latent profile (November 2020)			
	Acritical bystander	Acritical actor	Precritical bystander	Liberated actor
Acritical bystander ($n = 113$)	90%	10%	0.0%	0.0%
Acritical actor ($n = 44$)	15%	82%	3%	0.0%
Precritical bystander ($n = 282$)	0.0%	2%	94%	4%
Liberated actor ($n = 144$)	0.0%	4%	11%	86%

Note. The n for each group is based on the probability of being in the profile according to the final latent profile transition analysis model. Values presented in bold indicate stability between profiles constrained to be equal across time points.

Figure 1*Critical Consciousness Latent Profiles Using Z Scores*

Note. See the online article for the color version of this figure.

Stability Patterns

Of the five most prevalent transition patterns, three transition patterns consisted of stability patterns, that is, patterns wherein participants were likely to stay in the same profile across all three time points. These patterns were termed *acritical bystander stability pattern*, *precritical bystander stability pattern*, and *liberated actor stability pattern*.

Growth Patterns

The two remaining transition patterns demonstrated CC development between T1 and T2. The first pattern, termed *acritical bystander to precritical bystander growth pattern*, consisted of participants who were identified as acritical bystanders at T1 and then transitioned to precritical bystanders in T2 and T3. The second transition pattern, termed *precritical bystander to liberated actor growth pattern*, consisted of participants who identified as precritical bystanders at T1 and then transitioned to liberated actors in T2 and T3.

Latent Profile Transition Analysis in Relation to Discrimination

In general, participants likely to have high levels of CC reported higher levels of discrimination than participants likely to have average levels of CC (see Table 3 and Appendix M). The discrimination mean for the liberated actor stability pattern was significantly higher than the mean for the precritical bystander stability pattern ($z = -0.15$ vs. $z = -0.31$; $\chi^2 = 83.219, p < .001$). The mean of those who were in the precritical bystanders to liberated actor growth pattern was significantly higher than those in the precritical bystander stability pattern ($z = -0.28$ vs. $z = -0.31$; $\chi^2 = 247.65, p < .001$). There were no significant differences in mean levels of discrimination by racial/ethnic group (Appendix N).

Latent Profile Transition Analysis With Well-Being Distal Outcomes

In general, people who indicated lower levels of CC development over time had, on average, lower anxiety and stress; there were few differences for hopefulness (see Table 3 for a description of z scores and Appendix M for means). Baseline averages of anxiety, stress, and hopefulness for each CC pattern are also presented in Appendix M.

Anxiety

Table 3 displays the z scores and differences in anxiety by transition pattern. The acritical bystander stability pattern had, on average, lower levels of anxiety than the precritical bystander pattern at T2 ($z = -1.05$ vs. $z = -0.14$; $\chi^2 = 34.19, p < .001$) and T4 ($z = -0.92$ vs. $z = -0.17$; $\chi^2 = 17.95, p < .001$).

The acritical bystander stability pattern had, on average, lower levels of anxiety compared with the acritical to precritical growth pattern at T2 ($z = -1.05$ vs. $z = 0.96$; $\chi^2 = 107.57, p < .001$) and T4 ($z = -0.92$ vs. $z = 0.69$; $\chi^2 = 31.09, p < .001$).

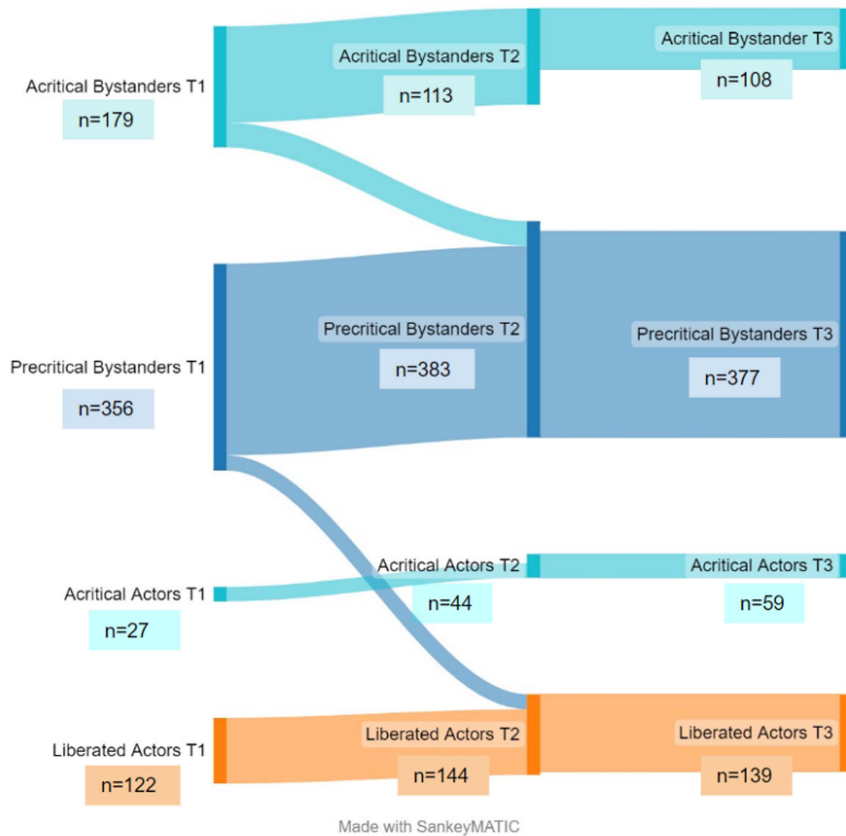
The precritical bystander stability pattern had, on average, lower levels of anxiety than the precritical to liberated growth pattern at T2 ($z = -0.14$ vs. $z = 0.96$; $\chi^2 = 37.50, p < .001$) and T4 ($z = -0.17$ vs. $z = 1.10$; $\chi^2 = 21.92, p < .001$). The other mean comparisons were not statistically significant.

Perceived Stress

Table 3 displays the z scores and differences in perceived stress by transition pattern. The acritical bystander stability pattern had lower levels of perceived stress, on average, than the precritical bystander stability pattern at T2 ($z = -0.91$ vs. $z = -0.04$; $\chi^2 = 30.19, p < .001$) and T4 ($z = -0.87$ vs. $z = -0.13$; $\chi^2 = 14.35, p < .001$).

Figure 2

Sankey Diagram Showing Movement Between the Five Common Critical Consciousness Transition Patterns Between T1 and T3



Note. Check Table 3 for percentages. *N* represents the number of people *most likely* to be in that profile. T = time. See the online article for the color version of this figure.

The acritical bystander stability pattern also had lower perceived stress, on average, than the acritical to precritical growth pattern at T2 ($z = -0.91$ vs. $z = 0.92$; $\chi^2 = 89.66$, $p < .00$) and T4 ($z = -0.87$ vs. $z = 0.70$; $\chi^2 = 24.38$, $p < .001$).

The precritical bystander stability pattern had lower perceived stress, on average, than the precritical to liberated growth pattern at T2 ($z = -0.04$ vs. $z = .94$; $\chi^2 = 26.62$, $p < .001$) and T4 ($z = -0.13$ vs. $z = 1.13$; $\chi^2 = 15.16$, $p < .001$). The other mean comparisons were not statistically significant.

Hopefulness

Table 3 displays the z scores and differences in hopefulness by transition pattern. The acritical to precritical growth pattern had higher levels of hopefulness agency, on average, than the acritical bystander stability pattern at T2 ($z = 0.86$ vs. $z = 0.31$; $\chi^2 = 9.82$, $p < .01$) and T4 ($z = 0.91$ vs. $z = 0.29$; $\chi^2 = 8.56$, $p < .01$).

The acritical bystander stability pattern participants had higher levels of hopefulness agency, on average, than the precritical bystander stability pattern at T2 ($z = 0.29$ vs. $z = -0.30$; $\chi^2 = 8.56$, $p < .01$).

Participants in the acritical bystander stability pattern had higher levels of hopefulness pathways, on average, than those in the

precritical bystander stability pattern at T2 ($z = 0.36$ vs. $z = -0.33$; $\chi^2 = 8.14$, $p < .01$). The other mean comparisons were not statistically significant.

Exploratory Latent Profile Transition Analysis With Sociodemographic Auxiliary Variables

See Table 3 for the probability, presented as odds ratios (OR), of being in a transition pattern by sociodemographic group. Participants in the acritical bystander stability pattern were more likely to be men rather than women, when compared with participants in the acritical bystander to precritical bystander growth pattern ($OR = 2.83$, CI [0.03, 6.63]) and the precritical bystander stability pattern ($OR = 3.83$, CI [1.12, 6.56]). Participants in the precritical bystanders to liberated actor growth pattern ($OR = 3.91$, CI [1.50, 6.33]) and participants in the liberated actor stability pattern ($OR = 3.41$, CI [1.10, 5.71]) were more likely to be LGBQ+ rather than straight when compared with participants in the precritical bystander stability pattern.

Participants who were white were more likely to be in the acritical bystander stability pattern than the precritical bystander stability pattern ($OR = 2.76$, CI [0.72, 4.79]). Participants who were white were also more likely to be in the liberated actor stability pattern than the precritical bystander stability pattern ($OR = 1.96$, CI [1.16,

Table 3
Association Between Well-Being Outcomes and Discrimination and Transition Patterns Calculated Using the Three-Step Process for Relating Latent Transition Analysis Models to Auxiliary Variables

Measure	Acritical bystander stability pattern	Acritical bystander to precritical bystander growth pattern	Precritical bystander stability pattern	Precritical bystander to liberated actor growth pattern	Liberated actor stability pattern
<i>Z score showing the association between continuous auxiliary variables and transition patterns</i>					
Stress					
Time 2	-0.91 _{ab}	0.92 _a	-0.04 _{bc}	0.94 _c	0.16
Time 4	-0.87 _{ab}	0.70 _a	-0.13 _{bc}	1.13 _c	-0.20
Anxiety					
Time 2	-1.05 _{ab}	0.96 _a	-0.14 _{bc}	0.96 _c	-0.16
Time 4	-0.92 _{ab}	0.69 _a	-0.17 _{bc}	1.10 _c	0.07
Hope agency					
Time 2	0.31 _{ab}	0.86 _a	-0.29 _b	0.67	-0.10
Time 4	0.29 _{ab}	0.91 _a	-0.30 _b	0.63	-0.05
Hope pathways					
Time 2	0.36 _a	0.80	-0.33 _a	0.55	-0.04
Discrimination					
Time 4	0.29	0.87	-0.31	0.60	0.02
Time 1	-0.42	-0.49 _a	-0.31 _{ab}	-0.28	-0.15 _b
<i>Sociodemographics (predicted probability from results of binomial/multinomial logistic regression relating each sociodemographic variable to a transition pattern)</i>					
Gender					
Women	0.34 _{ab}	0.60 _a	0.66 _b	0.54	0.86
Men	0.66 _{ab}	0.40 _a	0.34 _b	0.45	0.14
Sexuality					
LGBQ+	0.04	0.04	0.27 _{ab}	0.83 _a	0.57 _b
Straight	0.96	0.96	0.73 _{ab}	0.17 _a	0.43 _b
Race/ethnicity					
Asian	0.09	0.23	0.25 _a	0.02	0.13 _a
Black	0.02	0.00	0.05	0.00	0.00
Latinx	0.05	0.07	0.10	0.00	0.11
Multiple racial/ethnic groups and Middle Eastern/North African	0.13 _a	0.08	0.12 _a	0.10	0.10
White	0.71 _a	0.65	0.48 _{ab}	0.87	0.67 _b

Note. Results for well-being and discrimination variables used Wald tests to examine significance at $p < .05$ after Holm-Bonferroni corrections. Significant differences are indicated with subscript letters to indicate differences. LGBQ+ = lesbian, gay, bisexual, queer/questioning, and other minoritized identities.

5.88]). Participants who were Asian were more likely to be in the precritical bystander stability pattern than the liberated actor stability group ($OR = 2.37$, $CI [0.20, 4.54]$). Participants who identified with multiple races and/or Middle Eastern/North African were more likely to be in the acritical bystander stability pattern than the acritical bystander to precritical bystander growth pattern ($OR = 1.19$, $CI [0.05, 0.223]$). There were no significant differences in the probability of Black and Latinx participants being in the different transition patterns, and they overall had low probabilities of being in any of the major transition patterns than participants of the other racial/ethnic groups.

Discussion

The year 2020 saw a confluence of sociopolitical events that may have influenced the CC development of emerging adult college students (Moffitt et al., 2024; Quiles et al., 2023). Using a longitudinal approach to analyze CC development between April and November 2020, we identified three CC stability patterns and two CC transition patterns. Participants who maintained or increased their CC experienced more discrimination at T1 and more anxiety and stress as and after they developed their CC. Below, we situate our results within our study hypotheses and literature on CC development.

Identifying Critical Consciousness Latent Profile Transition Patterns

We identified the three hypothesized profiles (i.e., acritical bystanders, precritical bystanders, and liberated actors; Briggs et al., 2023; Godfrey et al., 2019; Schwarzenenthal et al., 2024) and a fourth profile, the acritical actors. We found that most youth remained in the same CC status between April and November 2020; however, some youth experienced growth in their CC (i.e., transitions from an earlier stage of SPD to a later stage). Few youth transitioned from later to earlier statuses over this ~6–7-month period, which suggests that emerging adults' CC was sustained or supported to grow. Particularly in college settings, emerging adults may have had access to opportunities to engage in CC development as they made sense of sociopolitical events, including through coursework and civic engagement (Maker Castro et al., 2024).

Although longitudinal patterns involving acritical actors were too infrequent to allow for transition analysis, we identified this understudied cross-sectional profile of participants taking action without corresponding reflection and motivation, which may help illuminate variation in SPD. This profile echoes Briggs et al.'s (2023) "precritical actor" profile and reflects youth engaged in action without considering the systemic issues being addressed. We may have captured a distinct starting point for CC development in that some youth may engage in action as an entryway to building their reflection and motivation; future research should further explore the developmental processes of these youth.

Relationships Between Transition Patterns and Discrimination

In line with our hypothesis, we found that levels of discrimination were higher among youth at more advanced stages of SPD, which may reflect the contribution that experiences of discrimination can have to CC development, including for those who are marginalized by

their race/ethnicity, gender, and/or sexuality (Anyiwo et al., 2018; Conner et al., 2023; Fine et al., 2018; Quiles et al., 2023). However, discrimination z scores were all below average in the five most prevalent transition patterns, suggesting that heightened levels of discrimination may be associated with less common longitudinal patterns of CC; future studies should explore these links.

Relationships Between Transition Patterns and Well-Being

In line with our hypotheses, we found that youth in the acritical bystander stability pattern had the lowest levels of both stress and anxiety and, in particular, their stress and anxiety were lower than both those who remained precritical bystanders and those who grew from acritical bystander to precritical bystander. Thus, we see that precritical youth with similarly low levels of critical action and critical motivation, but with a higher level of critical reflection, may be experiencing more stress and anxiety than their less conscious peers (Godfrey et al., 2019; Schwarzenenthal et al., 2024). Opportunity structures, or structured spaces like classrooms and youth programs, can support youth's SPD (Watts & Flanagan, 2007). Youth who could not access opportunity structures, especially during the early months of the COVID-19 pandemic when in-person activities were limited, may have felt unable to take action despite their burgeoning awareness of social injustice. The confluence of growing reflection and stymied action may have resulted in the higher levels of stress and anxiety we observed among those who were, and became, precritical bystanders.

Further in line with our hypotheses, we found that those who transitioned from precritical bystander to liberated actor had higher stress and anxiety than those who stayed as precritical bystanders. Here, youth's increasing sociopolitical involvement may heighten stress and anxiety (Hope et al., 2018). Our results highlight the need to consider well-being as a fundamental element of CC and to find ways, particularly via opportunity structures, to support youth who are learning and moving to engage with sociopolitical oppression (French et al., 2020; Watts & Flanagan, 2007). Notably, those who remained liberated actors throughout did not have the highest levels of stress and anxiety suggesting that, at least for some, CC may help heal from discrimination that engenders well-being (French et al., 2020).

Our hypothesis that those with consistently high levels of CC would experience more hopefulness (Christens et al., 2018) was not supported. Rather, the hope findings were more mixed. With hope agency, those who were acritical bystanders had higher levels than those who were precritical bystanders, yet growth from acritical bystander to precritical bystander (growth in critical reflection) was also associated with high levels of hope agency. Present measures of hope may capture both a "naïve" hopefulness from the lack of awareness of inequities and hopefulness from understanding that inequality can have structural roots (Suzuki et al., 2023). Future work should examine further which components of hope are relevant to CC.

Relationships Between Transition Patterns and Demographic Groups

We explored demographic differences without hypotheses, and as such these results must be interpreted with caution. The experiences of some groups were not explored due to aggregation into larger categories (e.g., we did not examine the unique experiences of

different subgroups of Asian youth), and some of our collapsed categories (i.e., Black and Latinx participants) likely still lacked statistical power due to small sample sizes. One statistically significant finding was of heterogeneity within white youth in their CC during this developmental period. White youth were both more likely to have relatively high level and relatively low levels of CC throughout the study period. These results suggest that white youth may not have developed CC development in response to the sociopolitical events of 2020 (e.g., BLM protests). For those who remained low in CC, perhaps they did not engage with their positions of racial privilege (Moffitt et al., 2024; Quiles et al., 2023) and/or they learned to condemn interpersonal racist incidents without implicating the power structures of white supremacy (Yi et al., 2023). Those already high in CC may have developed CC either despite their racial privilege (perhaps from other recent sociopolitical events like the 2016 election of Donald Trump; Dunn et al., 2022) or because of experiences of marginalization across other axes (e.g., gender, sexuality; Conner et al., 2023).

Results by gender showed that young men were more prevalent in the acritical bystander stability pattern than the SPD stage after it or in patterns where there was growth away from acritical bystander. This suggests that a position of gender privilege may hinder CC development (Connell, 2023). In addition, LGBTQ+ participants (compared with straight participants) were more prevalent in the lattermost SPD stages. Notably, LGBTQ+ participants were much more likely than their straight peers to experience growth into the liberated actor stage. Experiences with marginalization by sexuality may make LGBTQ+ youth responsive to opportunities to take action to challenge inequities.

Limitations and Future Directions

Our study has several limitations. Findings may not be generalizable to other time periods because of the distinct circumstances of the study period (e.g., global pandemic, surge in BLM protests). We encourage CC development research across sociohistorical contexts. Additionally, our research on how youth's social locations, including intersectionally, relate to CC development was limited by small sample sizes and the quantitative approach taken. Targeted sampling may increase statistical power for studying more specific subgroups' experiences, and qualitative approaches may better capture how CC is experienced differentially across social locations. For instance, youth with racial privilege may experience stress and anxiety differently from youth of color, who may gain awareness of how one's racial/ethnic group is subjected to historically rooted and ongoing racial trauma (Wray-Lake et al., 2023). In a second example, CC development among LGBTQ+ emerging adults is likely distinct based on race/ethnicity and gender (Vaccaro & Mena, 2011). Future work should also examine CC development among transgender and gender-diverse youth who may develop CC given their experiences challenging gendered systems (Fine et al., 2018) and CC development within specific youth spaces (like schools) to understand the role of contextual influences (Schwarzenenthal et al., 2024).

Conclusion

CC has long been hailed as a cornerstone of positive development for young people, especially those facing marginalization. However, there are "growing pains" to CC development: Our article found that

developing CC (i.e., moving from acritical bystander to precritical bystander, or from precritical bystander to liberated actor) was associated with higher levels of perceived stress and anxiety, as well as greater self-reported discrimination. Young people experiencing increasing awareness of, and motivation to disrupt, social inequalities may be in times of great personal and societal upheaval. We must provide these youth with the support needed for well-being and healing from oppression.

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(Appendices follow)

Appendix A

Details on Sociodemographic Information for Sample

At T1, participants self-reported their racial/ethnic background wherein multiple categories could be indicated; there was also a write-in option. We created five mutually exclusive ethnic/racial groups; due to small sample size, Middle Eastern/North African were included in the group with multiple races/ethnicities to create a multiracial/ethnic/other group.

At T1, participants were asked: “What is your gender identity?” Participants selected from seven options: *male or man*, *female or woman*, *trans male/trans man*, *trans female/trans woman*, *gender-queer/gender nonconforming*, *nonbinary*, or *different identity (please state)*. Self-reported gender identity at T1 was collapsed into two categories.

Participants reported their sexual orientation at T1. Participants were prompted: “Do you consider yourself to be:” and could choose

from the following options: *heterosexual or straight (i.e., not gay or lesbian)*, *gay or lesbian*, *bisexual*, *asexual*, *queer*, *pansexual*, *questioning*, or *another sexual identity (please state)*. Due to the limited sample size, we collapsed sexual orientation into two groups: heterosexual/straight youth and youth who identify as LGBTQ+ (i.e., bisexual, gay/lesbian, questioning, pansexual, asexual, or another sexual identity).

At T1, 27.6% of participants were in their first year of college, 25.9% were in their second year, 23.0% were in their third year, 22.7% were in their fourth year, and 1.0% reported other. Reported household incomes varied: 33.8% of participants reported household income <\$54,000 per year (i.e., 200% of the federal poverty line for a family of four in 2020), 27.9% reported household incomes between \$54,000 and \$99,999, and 32.2% reported household incomes >\$100,000.

Appendix B

Attrition Analysis

The study contained missing data due to loss to follow-up and partial survey completion (i.e., skipping some survey items). We examined whether those who were lost to follow-up had significantly different scores on study variables or demographics to those who remained in the study, using chi-square tests for dichotomous variables and *t* tests for continuous variables. We accounted for nonrandom attrition using two missing data approaches described in the analytic approach section.

Attrition rates were related to race/ethnicity, gender, and immigrant status, with Black youth, men, and nonimmigrant origin youth more likely to attrit at T2 compared with youth of other ethnic/racial groups,

immigrant-origin youth, and women and men continuing to be more likely to attrit (than women) at T3 and T4. Youth who remained in the study reported higher critical reflection at T2 and T3 than those who did not choose to participate—T2: $t(168.16) = -2.33, p < .05$; T3: $t(220.62) = -1.20, p < .05$. Youth who remained in the study at T2 reported more experiences with discrimination than those who did not participate $t(230.3) = 2.78, p < .01$. Youth who remained in the study at T3 demonstrated higher anxiety than those who did not choose to participate $t(191.04) = -2.04, p < .05$. No other differences in attrition emerged by demographic factors, critical motivation, critical action, or hopefulness.

Appendix C

Description of Evaluation of Latent Profile Analysis

To evaluate the models, we used statistical and substantive considerations. Statistically, the model fit indices included the Bayesian information criteria (BIC; Schwarz, 1978), the sample size adjusted BIC (Sclove, 1987), and the Akaike information criteria (Akaike, 1974), with the lowest observed values suggesting the best fit (although sometimes these indices do not reach a minimum; S. K. Johnson, 2021). To compare models,

we used Lo et al.’s (2001) likelihood ratio test, with a *p* value of >.05 indicating that the current model is a better fit over the model with fewer profiles. We assessed each model’s entropy with higher numbers indicating better class separation (Masyn, 2013). We also substantively examined the profiles to assess alignment with theoretical expectations and the distinctness of profiles (S. K. Johnson, 2021).

(Appendices continue)

Appendix D
Item-Level Descriptive Statistics for Study Variables

Measure	Item	<i>n</i>	<i>M</i>	<i>SD</i>	Median	Skew	Kurtosis	<i>SE</i>
Critical reflection April 2020 (T1)	Critical reflection 1	684.00	4.46	1.37	5.00	−0.85	0.16	0.05
	Critical reflection 2	684.00	4.50	1.39	5.00	−0.91	0.25	0.05
	Critical reflection 3	684.00	4.22	1.40	4.00	−0.66	−0.16	0.05
	Critical reflection 4	684.00	4.87	1.29	5.00	−1.21	0.95	0.05
Critical motivation April 2020 (T1)	Critical motivation 1	684.00	5.58	0.72	6.00	−2.02	4.96	0.03
	Critical motivation 2	684.00	5.31	1.06	6.00	−1.84	3.47	0.04
	Critical motivation 3	684.00	5.01	1.06	5.00	−1.10	1.15	0.04
	Critical motivation 4	684.00	5.00	1.09	5.00	−1.01	0.68	0.04
Critical action April 2020 (T1)	Critical action 1	684.00	1.78	1.14	1.00	1.34	0.68	0.04
	Critical action 2	684.00	1.92	1.24	1.00	1.17	0.17	0.05
	Critical action 3	684.00	1.61	0.91	1.00	1.57	2.11	0.03
	Critical action 4	684.00	1.60	0.90	1.00	1.67	2.64	0.03
	Critical action 5	684.00	1.83	1.16	1.00	1.31	0.66	0.04
Critical reflection July 2020 (T2)	Critical reflection 1	498.00	4.91	1.29	5.00	−1.34	1.58	0.06
	Critical reflection 2	498.00	4.95	1.28	5.00	−1.38	1.63	0.06
	Critical reflection 3	498.00	4.61	1.36	5.00	−0.99	0.49	0.06
	Critical reflection 4	498.00	5.09	1.21	5.50	−1.58	2.35	0.05
Critical motivation July 2020 (T2)	Critical motivation 1	498.00	5.57	0.80	6.00	−2.47	7.85	0.04
	Critical motivation 2	498.00	5.43	1.02	6.00	−2.24	5.31	0.05
	Critical motivation 3	498.00	5.11	1.08	5.00	−1.32	1.77	0.05
	Critical motivation 4	498.00	5.16	1.05	5.00	−1.42	2.12	0.05
Critical action July 2020 (T2)	Critical action 1	498.00	1.99	1.20	2.00	0.95	−0.25	0.05
	Critical action 2	498.00	1.92	1.24	1.00	1.15	0.09	0.06
	Critical action 3	498.00	2.03	1.17	2.00	0.88	−0.25	0.05
	Critical action 4	498.00	1.75	1.02	1.00	1.30	0.86	0.05
	Critical action 5	498.00	1.91	1.22	1.00	1.13	0.03	0.05
Critical reflection November 2020 (T3)	Critical reflection 1	409.00	4.86	1.27	5.00	−1.22	1.16	0.06
	Critical reflection 2	409.00	4.89	1.28	5.00	−1.29	1.27	0.06
	Critical reflection 3	409.00	4.67	1.31	5.00	−0.98	0.43	0.06
	Critical reflection 4	409.00	5.14	1.12	6.00	−1.43	1.86	0.06
Critical motivation November 2020 (T3)	Critical motivation 1	409.00	5.45	0.91	6.00	−2.00	4.35	0.04
	Critical motivation 2	409.00	5.34	1.06	6.00	−1.92	3.74	0.05
	Critical motivation 3	409.00	5.02	1.10	5.00	−1.34	2.10	0.05
	Critical motivation 4	409.00	5.16	1.03	5.00	−1.40	2.06	0.05
Critical action November 2020 (T3)	Critical action 1	409.00	2.01	1.20	2.00	0.94	−0.21	0.06
	Critical action 2	409.00	2.01	1.25	2.00	1.00	−0.19	0.06
	Critical action 3	409.00	1.90	1.11	1.00	0.97	−0.14	0.05
	Critical action 4	409.00	1.76	0.98	1.00	1.16	0.48	0.05
	Critical action 5	409.00	1.87	1.15	1.00	1.13	0.14	0.06
Anxiety July 2020 (T2)	Anxiety 1	504.00	1.62	0.94	2.00	−0.19	−0.86	0.04
	Anxiety 2	504.00	1.44	1.02	2.00	−0.01	−1.14	0.05
	Anxiety 3	504.00	1.56	1.02	2.00	−0.14	−1.11	0.05
	Anxiety 4	504.00	1.42	1.00	1.00	0.09	−1.07	0.04
	Anxiety 5	504.00	1.12	1.00	1.00	0.35	−1.06	0.04
	Anxiety 6	504.00	1.55	0.98	2.00	−0.14	−0.98	0.04
	Anxiety 7	504.00	1.27	1.02	1.00	0.24	−1.10	0.05
Anxiety July 2021 (T4)	Anxiety 1	316.00	1.75	0.82	2.00	−0.15	−0.58	0.05
	Anxiety 2	316.00	1.50	0.94	2.00	−0.01	−0.91	0.05
	Anxiety 3	316.00	1.65	0.93	2.00	−0.16	−0.84	0.05
	Anxiety 4	316.00	1.52	0.97	2.00	−0.04	−1.00	0.05
	Anxiety 5	316.00	1.24	1.01	1.00	0.23	−1.09	0.06
	Anxiety 6	316.00	1.58	0.93	2.00	−0.11	−0.85	0.05
	Anxiety 7	316.00	1.25	1.04	1.00	0.27	−1.12	0.06
Stress July 2020 (T2)	Stress 1	504.00	2.25	1.06	2.00	−0.19	−0.37	0.05
	Stress 2	504.00	2.40	1.17	3.00	−0.41	−0.62	0.05
	Stress 3	504.00	2.71	1.11	3.00	−0.58	−0.38	0.05
	Stress 4	504.00	2.33	0.94	2.00	−0.10	−0.18	0.04
	Stress 5	504.00	1.98	0.87	2.00	−0.07	−0.05	0.04
	Stress 6	504.00	1.89	1.10	2.00	0.07	−0.62	0.05
	Stress 7	504.00	2.23	0.92	2.00	−0.16	−0.11	0.04
	Stress 8	504.00	2.17	0.96	2.00	−0.08	−0.19	0.04
	Stress 9	504.00	2.36	1.12	2.00	−0.17	−0.71	0.05
	Stress 10	504.00	1.94	1.15	2.00	0.08	−0.79	0.05

(Appendices continue)

Appendix D (continued)

Measure	Item	<i>n</i>	<i>M</i>	<i>SD</i>	Median	Skew	Kurtosis	<i>SE</i>
Stress July 2021 (T4)	Stress 1	319.00	2.25	1.21	3.00	−0.27	−1.35	0.07
	Stress 2	319.00	2.14	1.30	3.00	−0.04	−1.47	0.07
	Stress 4	319.00	1.31	1.03	1.00	0.76	−0.59	0.06
	Stress 5	319.00	1.50	1.06	1.00	0.56	−1.06	0.06
	Stress 6	319.00	2.13	1.29	3.00	−0.15	−1.42	0.07
	Stress 7	319.00	2.40	1.14	3.00	−0.40	−1.24	0.06
	Stress 8	319.00	1.54	1.04	1.00	0.63	−1.01	0.06
	Stress 9	319.00	2.13	1.28	3.00	−0.04	−1.46	0.07
	Stress 10	319.00	1.89	1.35	1.00	0.16	−1.44	0.08
Hope_pathways July 2020 (T2)	Hope 1	504.00	6.03	1.32	6.00	−0.78	1.22	0.06
	Hope 3	504.00	6.37	1.28	7.00	−0.87	1.39	0.06
	Hope 4	504.00	6.15	1.33	6.00	−0.71	0.50	0.06
	Hope 5	504.00	5.92	1.42	6.00	−0.59	0.15	0.06
	Hope 2	504.00	6.26	1.42	6.00	−0.88	0.91	0.06
Hope_agency July 2020 (T2)	Hope 6	504.00	6.23	1.56	7.00	−1.09	0.97	0.07
	Hope 7	504.00	6.23	1.40	6.00	−0.85	0.77	0.06
	Hope 8	504.00	6.14	1.29	6.00	−0.51	−0.15	0.06
	Hope 1	321.00	6.28	1.29	7.00	−0.95	1.33	0.07
Hope_pathways July 2021 (T4)	Hope 3	321.00	6.52	1.18	7.00	−0.55	−0.05	0.07
	Hope 4	321.00	6.26	1.29	6.00	−0.81	0.70	0.07
	Hope 5	321.00	5.99	1.35	6.00	−0.61	0.40	0.08
	Hope 2	321.00	6.16	1.40	6.00	−0.74	0.39	0.08
Hope_agency July 2021 (T4)	Hope 6	321.00	6.22	1.54	6.00	−1.06	1.14	0.09
	Hope 7	321.00	6.30	1.39	7.00	−1.01	1.35	0.08
	Hope 8	321.00	6.13	1.38	6.00	−0.81	0.70	0.08
	Discrimination 1	684.00	0.57	0.97	0.00	1.75	2.40	0.04
Discrimination April 2020 (T1)	Discrimination 2	684.00	0.57	0.98	0.00	1.74	2.21	0.04
	Discrimination 3	684.00	0.70	1.10	0.00	1.38	0.73	0.04
	Discrimination 4	684.00	0.76	1.15	0.00	1.34	0.61	0.04
	Discrimination 5	684.00	0.29	0.74	0.00	2.94	8.97	0.03
	Discrimination 6	684.00	0.22	0.64	0.00	3.46	13.01	0.02
	Discrimination 7	684.00	0.22	0.64	0.00	3.46	13.01	0.02

Note. *SE* = standard error; T = time.

Appendix E**Longitudinal Measurement Invariance of Critical Consciousness Scale**

An initial longitudinal measurement invariance model with all items achieved configural and metric invariance (see Appendix Table E1) but did not achieve scalar invariance due to Item 11 (“Contacted a public official by phone, mail, or email to tell him/her how you felt about a particular social or political issue”). Modification indices indicated that this item’s loadings varied over time. Thus, Item 11 was removed, and longitudinal measurement invariance was rerun. The new longitudinal measurement invariance model without Item 11

achieved configural and metric invariance (see Appendix Table E2). Item 5 (“It is important for young people to know what is going on in the world”), which modification indices suggested to free, was freed to achieve partial metric invariance using the criteria that the change in comparative fit index is less than 0.01. Given the small difference in the intercept for Item 5 across time points, and the minimal comparative fit index adjustment (i.e., 0.937–0.939), final scale scores included Item 5.

Table E1
Longitudinal Measurement Invariance With All Items

Model	χ^2 (<i>df</i>)	CFI	Change in CFI	Decision
Configural invariance	1221.383 (628)	.949		Accept
Metric invariance	1256.316 (648)	.947	0.002	Accept
Scalar invariance	1458.746 (674)	.932	0.015	Reject

Note. CFI = comparative fit index.

(Appendices continue)

Table E2
Longitudinal Measurement Invariance Without Item 11

Model	χ^2 (df)	CFI	Change in CFI	Decision
Configural invariance	1071.752 (523)	0.948		Accept
Metric invariance	1089.860 (541)	0.948	0.000	Accept
Scalar invariance	1230.991 (565)	0.937	0.011	Reject
Partial scalar invariance (Item 5 freed)	1211.903 (564)	0.939	0.009	Accept

Note. CFI = comparative fit index.

Appendix F

Construct-Level Descriptive Statistics for Study Variables

Construct	Time	<i>M</i>	<i>S</i>	<i>SD</i>	Skew	Kurtosis	Skew to <i>SE</i> of skew ratio
Critical reflection	Time 1	4.51	0.05	1.22	−0.83	0.25	−8.94
	Time 2	4.89	0.05	1.18	−1.30	1.55	−12.05
	Time 3	4.89	0.06	1.14	−1.18	1.22	13.02
Critical motivation	Time 1	5.22	0.03	0.77	−1.12	1.62	−11.89
	Time 2	5.31	0.04	0.82	−1.52	2.72	−13.95
	Time 3	5.24	0.04	0.85	−1.42	2.30	9.14
Critical action	Time 1	1.78	0.03	0.91	1.21	0.67	−9.8
	Time 2	1.89	0.04	0.95	1.00	0.08	−11.85
	Time 3	1.91	0.05	0.98	0.95	0.01	7.93
Anxiety	Time 2	9.65	0.26	5.95	0.00	−0.90	−0.03
	Time 4	10.49	0.30	5.35	0.01	−0.75	0.05
Stress	Time 2	20.16	0.35	8.02	−0.51	0.11	−4.79
	Time 4	17.30	0.25	4.42	0.07	−0.71	0.51
Hopefulness pathways	Time 2	24.47	0.20	4.43	−0.56	0.74	−5.21
	Time 4	25.04	0.24	4.24	−0.61	0.33	−4.49
Hopefulness agency	Time 2	24.86	0.21	4.65	−0.60	0.09	−5.54
	Time 4	24.81	0.27	4.79	−0.76	0.71	−5.12
Discrimination	Time 1	0.52	0.03	0.72	1.73	3.11	18.51

Note. *SE* = standard error.

(Appendices continue)

Appendix G
Correlations With Confidence Intervals for Construct Variables

Variable	1	2	3	4	5	6	7	8	9
1. Critical reflection 1	—								
2. Critical reflection 2	.64** [.58, .69]	—							
3. Critical reflection 3	.69** [.63, .74]	.77** [.72, .80]	—						
4. Critical motivation 1	.53** [.47, .58]	.41** [.33, .48]	.46** [.38, .53]	—					
5. Critical motivation 2	.44** [.36, .51]	.61** [.55, .66]	.54** [.46, .61]	.48** [.41, .55]	—				
6. Critical motivation 3	.46** [.38, .54]	.49** [.41, .57]	.68** [.62, .72]	.52** [.45, .59]	.64** [.58, .70]	—			
7. Critical action 1	.26** [.19, .33]	.22** [.13, .30]	.24** [.15, .33]	.19** [.12, .26]	.21** [.12, .29]	.18** [.08, .27]	—		
8. Critical action 2	.27** [.19, .35]	.24** [.16, .32]	.21** [.11, .31]	.19** [.11, .28]	.16** [.09, .24]	.12** [.02, .22]	.68** [.62, .73]	—	
9. Critical action 3	.25** [.16, .34]	.23** [.13, .33]	.22** [.13, .31]	.24** [.14, .33]	.19** [.09, .29]	.14** [.04, .23]	.24** [.15, .32]	.21** [.10, .32]	—
10. Stress 2	.30** [.22, .38]	.30** [.22, .38]	.24** [.14, .33]	.18** [.10, .26]	.24** [.16, .32]	.17** [.07, .27]	.13** [.02, .23]	.28** [.20, .36]	.21** [.11, .31]
11. Stress 4	.23** [.13, .33]	.19** [.07, .30]	.18** [.06, .30]	.19** [.10, .27]	.25** [.17, .33]	.22** [.12, .31]	.20** [.11, .28]	.29** [.18, .40]	.18** [.06, .30]
12. Anxiety 2	.29** [.21, .37]	.16** [.05, .27]	.24** [.14, .34]	.15** [.04, .25]	.08 [−.04, .20]	.03 [−.07, .14]	.02 [−.09, .13]	.03 [−.09, .14]	.06 [−.06, .18]
13. Anxiety 4	.19** [.08, .29]	.16** [.05, .27]	.08 [−.04, .20]	.00 [−.09, .08]	.16** [.08, .25]	.07 [−.05, .19]	.02 [−.09, .13]	.03 [−.09, .14]	.06 [−.06, .18]
14. Hope-P 2	−.12** [−.20, −.03]	−.02 [−.11, .07]	−.08 [−.18, .02]	−.00 [−.09, .08]	.08 [−.04, .20]	.03 [−.07, .14]	.02 [−.09, .13]	.03 [−.09, .14]	.06 [−.06, .18]
15. Hope-P 4	−.20** [−.30, .09]	−.08 [−.19, .04]	−.04 [−.16, .08]	−.07 [−.18, .04]	.08 [−.04, .20]	.07 [−.05, .19]	.02 [−.09, .13]	.03 [−.09, .14]	.06 [−.06, .18]
16. Hope-A 2	−.13** [−.22, −.04]	.00 [−.09, .09]	−.02 [−.12, .08]	−.00 [−.09, .08]	.18** [.10, .27]	.12** [.02, .22]	−.03 [−.12, .06]	−.11** [−.19, −.02]	−.03 [−.13, .07]
17. Hope-A 4	−.17** [−.27, −.06]	−.04 [−.16, .08]	−.06 [−.18, .06]	−.03 [−.14, .08]	.13** [.01, .24]	.11 [−.01, .23]	−.07 [−.17, .04]	−.04 [−.15, .08]	.01 [−.11, .13]
18. Discrimination	.02 [−.05, .10]	−.02 [−.11, .07]	−.01 [−.10, .09]	−.07 [−.14, .01]	−.06 [−.14, .03]	−.12** [−.22, −.03]	.17** [.10, .24]	.20** [.11, .28]	.25** [.16, .34]
Variable	10	11	12	13	14	15	16	17	18
10. Stress 2	—								
11. Stress 4	.46** [.36, .54]	—							
12. Anxiety 2	.79** [.75, .82]	.50** [.41, .59]	—						
13. Anxiety 4	.44** [.34, .53]	.67** [.60, .72]	.57** [.49, .64]	—					
14. Hope-P 2	−.35** [−.42, −.27]	−.30** [−.40, −.19]	−.21** [−.29, −.13]	−.24** [−.35, −.13]	—				
15. Hope-P 4	−.19** [−.30, −.08]	−.24** [−.34, −.14]	−.13** [−.24, −.01]	−.19** [−.29, −.08]	.61** [.53, .68]	—			
16. Hope-A 2	−.39** [−.46, −.31]	−.30** [−.40, −.19]	−.22** [−.30, −.14]	−.26** [−.37, −.15]	.74** [.70, .78]	.47** [.37, .55]	—		
17. Hope-A 4	−.22** [−.33, −.11]	−.37** [−.47, −.28]	−.17** [−.28, −.06]	−.25** [−.35, −.15]	.48** [.38, .56]	.69** [.63, .74]	.62** [.55, .69]	—	
18. Discrimination	.11** [.03, .19]	.17** [.06, .27]	.14** [.05, .22]	.17** [.06, .27]	−.05 [−.14, .04]	.03 [−.08, .14]	−.03 [−.12, .06]	.01 [−.10, .12]	—

Note. Values in square brackets indicate the 95% confidence interval for each correlation. The confidence interval is a plausible range of population correlations that could have caused the sample correlation. Values presented in bold indicate significant correlations. P = pathways; A = agency.

* $p < .05$. ** $p < .01$.

(Appendices continue)

Appendix H

Model Fit Indices for Latent Profiles of Critical Consciousness

Number of profiles	Smallest profile	LL	AIC	BIC	SABIC	LMR-LRT	Entropy
Time 1							
Variance-covariance structure type 1: class-invariant diagonal							
One		-2800.34	5612.68	5639.85	5620.8		
Two	202	-2651.57	5323.15	5368.43	5336.68	<0.001	0.733
Three	112	-2571.11	5170.22	5233.61	5189.16	0.0775	0.782
Four	29	-2492.34	5020.69	5102.19	5045.04	0.0002	0.826
Five	26	-2436.74	4917.47	5017.08	4947.23	0.1069	0.834
Six	39	-2407.86	4867.72	4985.45	4902.9	0.0491	0.835
Seven	27	-2388.23	4836.46	4972.29	4877.04	0.1361	0.798
Variance-covariance structure type 2: profile-varying diagonal							
Two	199	-2467.9	4961.81	5020.67	4979.39	0	0.852
Variance-covariance structure type 3: profile-invariant unrestricted							
One		-2661.61	5341.21	5381.97	5353.39		
Two	128	-2535.87	5097.74	5156.60	5115.33	<0.001	0.902
Three	59	-2485.85	5005.69	5082.67	5028.69	0.0187	0.884
Four	47	-2437.05	4916.10	5011.19	4944.513	0.0044	0.903
Five	46	-2400.72	4851.44	4964.64	4885.26	0.6421	0.871
Six	12	-2374.38	4806.766	4938.08	4845.1	0.4639	0.876
Seven	5	-2374.38	4774.20	4923.62	4818.84	0.1525	0.883
Time 2							
Variance-covariance structure type 1: class-invariant diagonal							
One		-2076.32	4164.64	4189.91	4170.86		
Two	108	-1923.52	3867.03	3909.14	3877.40	0.0042	0.829
Three	33	-1841.74	3711.48	3770.43	3725.99	0.0023	0.887
Four	26	-1793.70	3623.4	3699.19	3641.06	0.2338	0.901
Five	26	-1739.01	3522.02	3614.65	3544.83	0.0128	0.880
Six	15	-1701.41	3454.82	3564.29	3481.77	0.0647	0.889
Seven	15	-1679.90	3419.81	3546.13	3450.91	0.0841	0.865
Variance-covariance structure type 3: profile-invariant unrestricted							
One		-1947.109	3912.22	3950.11	3921.58		
Two	28	-1863.78	3753.56	3808.30	3767.04	<0.001	0.968
Three	29	-1779.88	3593.76	3665.34	3611.38	<0.001	0.912
Four	27	-1738.93	3519.86	3608.28	3541.62	0.1410	0.910
Five	24	-1711.106	3472.21	3577.48	3498.13	0.1029	0.887
Six	13	-1687.58	3433.15	3555.26	3463.21	0.4103	0.878
Seven	8	-1663.31	3392.62	3531.57	3426.83	0.0876	0.889
Time 3							
Variance-covariance structure type 1: class-invariant diagonal							
One		-1716.97	3445.94	3470.02	3450.98		
Two	77	-1564.79	3151.58	3191.72	3159.99	0.0006	0.886
Three	46	-1522.75	3073.50	3129.69	3085.27	0.4327	0.801
Four	6	-1480.55	2997.09	3069.34	3012.22	0.0722	0.893
Five	24	-1436.22	2916.44	3004.74	2934.93	0.0501	0.850
Six	9	-1405.77	2862.77	2967.13	2884.63	0.3693	0.874
Seven	9	-1386.04	2832.08	2952.49	2857.30	0.1294	0.851
Variance-covariance structure type 3: profile-invariant unrestricted							
One		-1582.2	3182.41	3218.53	3189.97		
Two	99	-1514.37	3054.75	3106.92	3065.67	<0.001	0.881
Three	28	-1474.41	2982.83	3051.06	2997.12	0.0903	0.880
Four	24	-1442.65	2927.30	3011.58	2944.95	0.0654	0.885
Five	14	-1413.98	2877.96	2978.3	2898.97	0.3160	0.903
Six	14	-1385.85	2829.69	2946.09	2854.07	0.2317	0.891
Seven	13	-1326.66	2791.31	2923.77	2819.05	0.0337	0.899

Note. $N = 684$. LL = log likelihood; AIC = Akaike information criterion; BIC = Bayesian information criterion; SABIC = sample size adjusted BIC; LMR-LRT = p value of the adjusted Lo et al. (2001) likelihood ratio test. Models that did not converge are not shown. Bolded rows indicate the profiles selected at wave.

(Appendices continue)

Appendix I

Classification Diagnostics for Each Selected Latent Profile Analysis Model

Class	Model-estimated posterior class probability	90% CI of model-estimated posterior class probability	Modal class assignment proportion	Average posterior class probability	Odds of correct classification
Four-profile class-invariant diagonal					
1	0.04	[0.03, 0.06]	0.04	0.92	263.49
2	0.16	[0.13, 0.20]	0.16	0.92	56.63
3	0.24	[0.18, 0.30]	0.23	0.86	19.81
4	0.56	[0.50, 0.62]	0.57	0.91	7.59
Four-profile profile-invariant unrestricted					
1	0.06	[0.04, 0.82]	0.06	0.93	204.51
2	0.70	[0.66, 0.74]	0.69	0.97	12.15
3	0.19	[0.15, 0.23]	0.19	0.91	131.79
4	0.05	[0.03, 0.07]	0.05	0.93	246.43
Four-profile profile-invariant unrestricted					
1	0.06	[0.04, 0.09]	0.06	0.85	80.51
2	0.17	[0.13, 0.22]	0.16	0.85	28.28
3	0.08	[0.06, 0.11]	0.09	0.97	303.10
4	0.68	[0.63, 0.73]	0.69	0.97	12.82

Note. CI = confidence interval.

Appendix J

Follow-Up Alternative Analysis of Latent Profile Analysis at T3 to Understand the Acritical Actors

We used the Bolck–Croon–Hagenaars procedure (Asparouhov & Muthén, 2014) to conduct cross-sectional analysis of the acritical actors at T3, when participants were most likely to be this profile according to the LPTA unconditional model results (see Table 2). We used Wald χ^2 tests to assess the significance of the relations between CC profiles, discrimination, and well-being variables.

The acritical actors reported the highest levels of discrimination at T1 with a mean of 1.04 compared with a mean of 0.22 for acritical bystanders ($\chi^2 = 15.90$, $p < .001$), mean of 0.40 for precritical bystanders ($\chi^2 = 3.54$, $p = .06$), and mean of 0.72 for liberated actors ($\chi^2 = 11.29$, $p < .01$).

At T2, acritical actors reported less perceived stress ($\chi^2 = 8.58$, $p < .01$) and anxiety ($\chi^2 = 11.65$, $p < .01$) than liberated actors. At T2, acritical actors also had less hopefulness pathways than acritical bystanders ($\chi^2 = 6.74$, $p < .01$). At T4, acritical actors had more anxiety than acritical bystanders

($\chi^2 = 16.44$, $p < .001$) and precritical bystanders ($\chi^2 = 20.70$, $p < .001$).

The results for the manual Bolck–Croon–Hagenaars procedure examining associations between T3 profiles and sociodemographics (i.e., gender, sexuality, and race/ethnicity) are presented in Appendix K. We focus our reporting on significant differences between participants in the acritical actor profile compared with other profiles. Participants in the acritical actor profile were more likely to be men than women, when compared with participants in the acritical bystander profile ($OR = 2.38$, $CI [0.68, 8.33]$). Participants in the acritical actor profile were more likely to be White rather than other racial/ethnic groups, when compared with participants in the precritical bystanders ($OR = 2.13$, $CI [0.85, 5.26]$). Participants in the acritical actor profile were more likely to be White ($OR = 16.67$, $CI [4.55, 50]$) or Black ($OR = 2.7$, $CI [0.74, 10]$) rather than other racial/ethnic groups, when compared with the liberated actors. No other odds ratios were significant when comparing acritical actors with other profiles.

(Appendices continue)

Appendix K

Cross-Sectional Latent Profile Analysis at Time 3 With Comparisons of Estimated Means of Well-Being Outcomes, Discrimination, and Probabilities of Membership by Critical Consciousness Profiles

Measure	Acritical bystander		Precritical bystander		Liberated actor		Acritical actor	
	<i>M</i>	<i>SE</i>	<i>M</i>	<i>SE</i>	<i>M</i>	<i>SE</i>	<i>M</i>	<i>SE</i>
Stress								
Time 2	18.96	1.44	20.14	1.18	24.06	1.16	19.30 _c	1.44
Time 4	16.47	1.28	16.78	0.33	19.00	0.83	18.40	0.96
Anxiety								
Time 2	8.95	1.13	9.58	0.38	13.19	0.87	8.80 _c	0.91
Time 4	8.06	1.27	9.92	0.40	12.04	1.08	14.35 _a	0.88
Hope agency								
Time 2	26.18	1.03	25.08	0.29	24.91	0.71	22.06	1.12
Time 4	26.49	1.09	24.90	0.33	25.56	0.82	23.41 _a	1.56
Hope pathways								
Time 2	26.29	0.87	24.46	0.28	24.46	0.28	22.79 _a	1.02
Time 4	27.79	0.76	24.85	0.32	26.04	0.75	24.32	1.33
Discrimination								
Time 1	0.22	0.08	0.40	0.12	0.72	1.18	0.14 _{ab}	0.04
Sociodemographics (predicted probability from results of binomial/multinomial logistic regression relating each sociodemographic variable to a profile)								
Gender								
Women	0.37	0.08	0.66	0.03	0.96	0.04	0.20	0.08
Men	0.63	0.08	0.34	0.03	0.04	0.04	0.80	0.08
Sexuality								
LGBQ+	0.00	0.00	0.20	0.03	0.80	0.06	0.15	0.07
Straight	1.00	0.00	0.80	0.03	0.21	0.06	0.86	0.7
Race/ethnicity								
Asian	0.12	0.06	0.29	0.03	0.03	0.03	0.18	0.07
Black	0.02	0.02	0.01	0.01	0.02	0.02	0.45	0.09
Latinx	0.06	0.04	0.12	0.02	0.01	0.02	0.09	0.05
Multiple races/ethnicities and Middle Eastern/North African	0.11	0.05	0.14	0.02	0.08	0.04	0.00	0.03
White	0.70	0.08	0.45	0.03	0.86	0.06	0.28	0.09

Note. *N* = 461. Values presented with subscript letters indicate significant differences. *SE* = standard error; LGBQ+ = lesbian, gay, bisexual, queer/questioning, and other minoritized identities.

Appendix L

Model Fit Statistics for Measurement Invariance Across Time for the Four-Profile Unconditional Model

Model	LL	AIC	BIC	SABIC	Entropy
Unconstrained model	−5392.024	10940.047	11293.228	11045.568	0.859
Constrained model with profiles constrained to be equal across time	−5426.340	10960.681	11205.190	11033.733	0.850

Note. Values in bold were compared across the two models. LL = log likelihood; AIC = Akaike information criterion; BIC = Bayesian information criterion; SABIC = sample size adjusted BIC.

(Appendices continue)

Appendix M
Mean Scores on Stress and Anxiety Outcomes by Transition

	Mean score on outcome variable (<i>SE</i> of <i>M</i>)											
	Anxiety (T1)	Anxiety (T2)	Anxiety (T4)	Stress (T1)	Stress (T2)	Stress (T4)	Hope-P (T1)	Hope-P (T2)	Hope-P (T4)	Hope-A (T1)	Hope-A (T2)	Hope-A (T4)
Transition												
Acritical bystander stability group	6.97 (0.78)	3.41 (0.57)	5.58 (0.72)	17.52 (1.60)	12.85 (0.97)	13.46 (0.71)	27.00 (0.53)	26.07 (0.94)	26.25 (0.91)	23.74 (0.31)	26.32 (0.89)	26.22 (0.90)
Acritical bystander to precritical bystander transition group	10.30 (1.31)	15.36 (0.95)	14.19 (1.33)	23.52 (1.35)	27.55 (1.20)	20.38 (1.17)	23.53 (1.25)	28.00 (1.03)	28.72 (0.73)	23.55 (1.23)	28.86 (0.99)	29.15 (0.77)
Precritical bystander stability group	10.52 (0.37)	8.81 (0.68)	9.56 (0.62)	23.18 (0.56)	19.86 (0.87)	16.74 (0.53)	25.15 (0.29)	23.01 (0.35)	23.72 (0.41)	25.65 (0.58)	23.48 (0.46)	23.35 (0.56)
Precritical bystander to Liberated actor transition group	15.12 (0.90)	15.36 (0.89)	16.40 (1.37)	28.98 (1.26)	27.70 (1.25)	22.28 (1.35)	25.026 (1.04)	28.93 (0.40)	27.60 (2.32)	25.01 (1.05)	27.98 (1.42)	27.85 (1.38)
Liberated actor stability group	14.37 (0.90)	10.09 (1.39)	9.63 (1.5)	27.40 (1.07)	21.45 (1.29)	16.40 (1.16)	25.58 (0.57)	24.31 (0.62)	25.12 (0.91)	24.28 (0.31)	24.38 (0.57)	24.55 (0.86)

Note. T = time; P = pathways; A = agency; *SE* = standard error.

Appendix N
Mean Scores of Discrimination by Racial/Ethnic Groups at T1

Racial/ethnic group	<i>M</i>	<i>SD</i>
Asian	0.57	0.70
Black	0.64	0.86
Latinx	0.40	0.60
Multiple racial/ethnic groups	0.53	0.80
White	0.48	0.67

Note. Mean discrimination scores are based on a scale from 0 to 4. The analysis of variance indicated no statistically significant differences in discrimination scores between the different racial groups, $F(1, 3) = 1.77, p = 0.13$. T = time.

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